EXECUTIVE REPORT & MARKET ANALYSIS

Quantitative Analysis: CS2 Assets vs. Bitcoin & S&P 500
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Timeframe: 2016–2025
Status: Finalized

1. Executive Summary

This study evaluates the historical risk, return, and volume/price dynamics of three distinct asset classes under a unified timeframe (2016–2025):

- **Traditional Financial Asset:** S&P 500 ETF (SPY)
- **Crypto Asset:** Bitcoin (BTC)
- **Alternative Digital Asset:** CS2 Gamma 2 Case (Steam Community Market)

The primary objective is to analyze how virtual goods in closed economies behave compared to regulated equity markets and decentralized cryptocurrencies when measured using standard quantitative financial metrics.

2. Key Analytical Findings

While the broader ETL pipeline processes monthly historical data starting in 2016, the comparative performance metrics below isolate the timeframe from **June 2021 to December 2025** to evaluate assets following the structural transition into a scarcity asset.

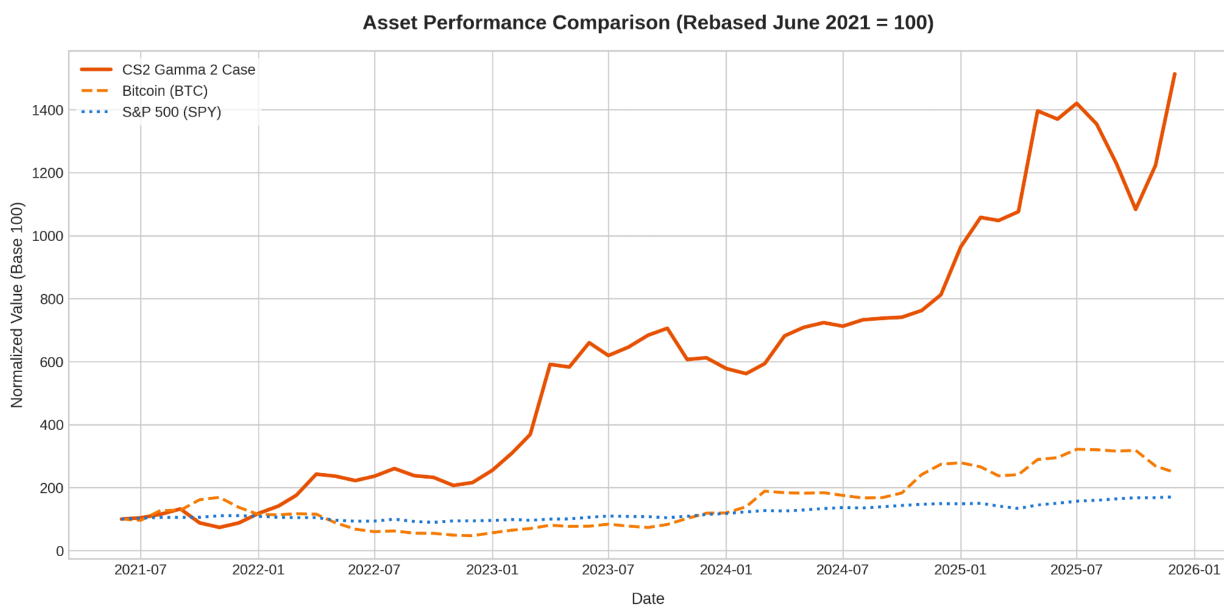
Asset	Total Return	Monthly Volatility	Asset Class Characteristics
CS2 Gamma 2 Case	+1,413.56%	22.63%	Alternative Asset (Illiquid, closed economy, limited supply)
Bitcoin (BTC)	+148.09%	19.97%	Cryptocurrency (High liquidity, 24/7 trading, global adoption)

Asset	Total Return	Monthly Volatility	Asset Class Characteristics
S&P 500 (SPY)	+70.86%	3.53%	Traditional Equities (Regulated, high liquidity, low volatility)

Structural Market Milestone (June 2021)

In June 2021, the Gamma 2 Case was moved to Counter-Strike's "Rare Drop Pool," drastically reducing new supply creation while consumption (case unboxings) remained active. This generated a structural supply deficit that drove exponential growth outside the macroeconomic cycle.

3. Performance Visualizations



4. Market Dynamics & Financial Outlook

- **Supply Shock & Price Inelasticity:** The transition to the rare drop pool triggered a clear decoupling from traditional markets, generating asymmetric returns driven by asset scarcity.

- **Risk / Return Profile:** While the virtual asset delivered substantially higher returns, it exhibited higher monthly volatility (22.63%) compared to Bitcoin (19.97%). Additionally, platform friction risks exist (15% Steam Community Market fee and capital extraction constraints).
- **Macroeconomic Correlation:** The S&P 500 offered risk-adjusted capital preservation (3.53% volatility), whereas CS2 assets displayed a correlation close to zero with equities, offering potential idiosyncratic diversification benefits.

5. Data Engineering & ETL Pipeline

- **SPY & BTC Data:** Daily price and volume history extracted using Python via the Yahoo Finance API (yfinance).
- **Steam Market Data:** Scraping and ingestion from Steam Community Market API endpoints (/market/pricehistory/), managing USD currency conversion and rate-limiting.
- **Processing & Normalization:** Handled stock market calendar disparities (traditional markets operating ~252 days/year vs. continuous 24/7 trading in Crypto and Steam) through temporal resampling and monthly aggregation.